1. FCFS Program

Input:

```
#include<stdio.h>

int main()
{
   int n;

   printf("Enter number of processes: ");
   scanf("%d",&n);

   int pid[n], at[n], bt[n], ct[n], tat[n], wt[n];

   printf("\nEnter Process ID, Arrival Time and Burst Time:\n");

   for(int i=0;i<n;i++)
   {
      scanf("%d %d %d",&pid[i],&at[i],&bt[i]);
   }

   // Sorting according to Arrival Time
   for(int i=0;i<n-1;i++)
   {
      for(int j=i+1;j<n;j++)
      {
         if(at[i] > at[j])
         {
            int temp;

            temp = at[i];
            at[i] = at[j];
            at[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;

            temp = pid[i];
            pid[i] = pid[j];
            pid[j] = temp;
         }
      }
   }

   int time = 0;
```

```
    for(int i=0;i<n;i++)
    {
        if(time < at[i])
            time = at[i];

        ct[i] = time + bt[i];
        time = ct[i];

        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];
    }

    float avgTAT = 0, avgWT = 0;

    printf("\nProcess  AT  BT  CT  TAT  WT\n");

    for(int i=0;i<n;i++)
    {
        printf("%5d %3d %3d %3d %4d %4d\n",
        pid[i],at[i],bt[i],ct[i],tat[i],wt[i]);

        avgTAT += tat[i];
        avgWT += wt[i];
    }

    avgTAT /= n;
    avgWT /= n;

    printf("\nAverage Turnaround Time = %.2f",avgTAT);
    printf("\nAverage Waiting Time = %.2f",avgWT);

    return 0;
}
```

Output:

```
Enter number of processes: 4

Enter Process ID, Arrival Time and Burst Time:
1 0 3
2 3 4
3 5 4
4 7 3

Process  AT  BT  CT  TAT  WT
    1    0   3   3    3   0
    2    3   4   7    4   0
    3    5   4  11    6   2
    4    7   3  14    7   4

Average Turnaround Time = 5.00
Average Waiting Time = 1.50
PS C:\Users\Parv Jain\OneDrive\Desktop\OS Lab>
```

2. Shortest time Job

Input:

```
#include<stdio.h>
#include<stdlib.h>
#define max(a,b) ((a>b)? (a):(b))

int main(){
   int a=4,b=6,temp,comp_count=0;
   int sjf[a][b];

   printf("Enter the Process, Arrival Time and Burst Time:\n");
   for(int i=0; i<a; i++){
      for(int j=0; j<3; j++){
         scanf("%d",&sjf[i][j]);
      }
   }

   for(int i=0; i<a-1; i++){
      for(int j=0; j<a-i-1; j++){
         if(sjf[j][1] > sjf[j+1][1]){
            for(int k=0; k<3; k++){
               temp = sjf[j][k];
               sjf[j][k] = sjf[j+1][k];
               sjf[j+1][k] = temp;
            }
         }
      }
   }

   int curr_time =0, is_comp[4] = {0};
   while(comp_count < a){
      int best_index = -1, min_burst = 9999;

      for(int i=0; i<a; i++){
         if(sjf[i][1] <= curr_time && is_comp[i] == 0){
            if(sjf[i][2] < min_burst) {
         min_burst = sjf[i][2];
         best_index = i;
      }
   }
}

if(best_index != -1){
   curr_time += sjf[best_index][2];
         sjf[best_index][3] = curr_time;
```

```
            sjf[best_index][4] = sjf[best_index][3] - sjf[best_index][1];
            sjf[best_index][5] = sjf[best_index][4] - sjf[best_index][2];
        is_comp[best_index] = 1;
        comp_count++;
    }
    else{
        curr_time++;
    }
}

printf("Process\tAT\tBT\tCT\tTAT\tWT\n");
    for(int i=0; i<a; i++){
        for(int j=0; j<b; j++){
            printf("%d\t",sjf[i][j]);
        }
        printf("\n");
    }

    int sumTAT =0, sumWT=0;
    for(int i=0; i<a; i++){
        sumTAT += sjf[i][4];
        sumWT += sjf[i][5];
    }
    printf("Average TAT: %2f",(float)sumTAT/a);
    printf("Average WT: %2f",(float)sumWT/a);
}
```

Output:

```
Enter the Process, Arrival Time and Burst Time:
1 0 3
2 3 4
3 6 5
4 4 3
Process AT      BT      CT      TAT     WT
1       0       3       3       3       0
2       3       4       7       4       0
4       4       3       10      6       3
3       6       5       15      9       4
Average TAT: 5.500000Average WT: 1.750000
PS C:\Users\Parv Jain\OneDrive\Desktop\OS Lab>
```

3. Shortest Remaining Time

Input:

```
#include <stdio.h>

int main() {
  int n, completed = 0, curr_time = 0, pid[10], at[10], bt[10], remt[10],
        ct[10], tat[10], wt[10], st[10], rt[10], i, j;
  double tat_sum = 0, wt_sum = 0;

  printf("Enter no. of processes: ");
  scanf("%d", &n);

  for (i = 0; i < n; i++) {
    pid[i] = i + 1;

    printf("Enter arrival time for process[%d]: ", i + 1);
    scanf("%d", &at[i]);

    printf("Enter burst time for process[%d]: ", i + 1);
    scanf("%d", &bt[i]);
    remt[i] = bt[i];
  }

  while (completed != n) {

    int min = 99999999,
        s = -1;

    for (i = 0; i < n; i++) {
      if (at[i] <= curr_time && remt[i] > 0 && remt[i] < min) {
        min = remt[i];
        s = i;
      }
    }

    if (s == -1) {
      curr_time++;
      continue;
    }

    if (remt[s] == bt[s])
      st[s] = curr_time;

    remt[s]--;

    if (remt[s] == 0) {

      ct[s] = curr_time + 1;
```

```
    tat[s] = ct[s] - at[s];
    tat_sum += tat[s];

    wt[s] = tat[s] - bt[s];
    wt_sum += wt[s];

    rt[s] = st[s] - at[s];

    completed++;
   }
   curr_time++;
  }

  printf("\nP\tAT\tBT\tCT\tTAT\tWT\tRT\n\n");

  for (j = 1; j < n + 1; j++) {
   for (i = 0; i < n; i++) {
    if (pid[i] == j)
     printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n", pid[i], at[i], bt[i], ct[i],
          tat[i], wt[i], rt[i]);
   }
  }

  printf("\nAverage TAT: %.2f\n", (tat_sum / n));
  printf("Average WT: %.2f\n", (wt_sum / n));
}
```

Output:

```
Enter no. of processes: 3
Enter arrival time for process[1]: 0
Enter burst time for process[1]: 3
Enter arrival time for process[2]: 2
Enter burst time for process[2]: 4
Enter arrival time for process[3]: 4
Enter burst time for process[3]: 3

P       AT      BT      CT      TAT     WT      RT

P1      0       3       3       3       0       0
P2      2       4       7       5       1       1
P3      4       3       10      6       3       3

Average TAT: 4.67
Average WT: 1.33
PS C:\Users\Parv Jain\OneDrive\Desktop\OS Lab>
```